

PhonePe Payment Insights Dashboard

Project Documentation

Power BI · DAX · Data Modeling · Power Query

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Dashboard Preview



1. Problem Statement

PhonePe processes millions of UPI transactions daily across services like recharges, bill payments, loans, and money transfers. As a Data Analyst supporting the product/business team, the goal of this project was to:

- Clean and structure a raw transactional dataset
- Build a reliable data model connecting user and transaction data
- Design KPIs and visuals that reveal growth trends, service performance, and user behavior
- Present findings in a way that's directly actionable for business decisions

2. Data Understanding

Two source tables were provided:

All_Users (~108K rows)

Column	Description
User_ID	Unique identifier for each user
Name	User's name
Age	User's age
Join_Date	Date the user joined PhonePe

All_Transactions (~300K rows)

Column	Description
Transaction_ID	Unique transaction identifier
Amount	Transaction amount
User_ID	Links to All_Users
Service	Top-level category (Loans, Insurance, Money Transfer, Recharge Bills)
Column	Description
Service Type	Sub-category (e.g. Bike Loan, DTH, FASTag Recharge, Mobile Recharge)
Payment_Status	Successful / Failed
Reason	Successful / Server error / Wrong PIN / Insufficient amount
Date	Transaction date (Jan 2024 - Dec 2024)

3. Data Cleaning (Power Query)

- Verified column headers and data types across both tables (Text, Number, Date, Decimal)
- Removed duplicate rows on **Transaction_ID** and **User_ID**
- Removed blank rows in All_Users (missing User_ID would have broken the relationship to All_Transactions)
- Standardized category labels (e.g. removed underscores: Recharge_Bills → Recharge Bills) for readability
- Used Power BI's **Column Quality** / **Column Distribution** profiling tools to catch blanks and inconsistent values before modeling

Why clean before modeling? Uncleaned data (duplicates, blanks, inconsistent labels) breaks relationships and produces misleading visuals — this is a standard first step before any analysis.

4. Data Modeling

Built a simple **star schema** with the Date table and All_Users both feeding into All_Transactions as the central fact table.

Relationship	Cardinality	Cross-filter
All_Users → All_Transactions	1-to-Many	Single
Date_Table → All_Transactions	1-to-Many	Single

Why 1-to-Many, not Many-to-Many? Each transaction belongs to exactly one user and happened on exactly one date — the 'many' side always sits on the transactions table. Accepting a Many-to-Many relationship (a common beginner mistake) would allow ambiguous filtering and incorrect aggregations.

Why single-direction cross filtering? Filters flow from the '1' side (Users, Date) down to the 'Many' side (Transactions) — matching how the dashboard is used. Bi-directional filtering was avoided since it wasn't needed and can cause ambiguous filter paths in larger models.

5. Custom Date Table (DAX)

Rather than relying on the transaction date column directly, a dedicated Date Table was created:

```
Date Table =
ADDCOLUMNS (
    CALENDAR(MIN(All Transactions[Date]), MAX(All Transactions[Date])),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month", FORMAT([Date], "MMM"),
    "Quarter", "Q" & FORMAT([Date], "Q"),
    "Week Day", FORMAT([Date], "DDDD"),
    "Day Number", WEEKDAY([Date], 2),
    "Weekend", IF(WEEKDAY([Date], 2) >= 6, "Weekend", "Weekday")
)
```

Why a separate date table?

- Enables clean time-intelligence functions (MoM, YTD, QTD)
- Avoids gaps if a date has zero transactions
- Lets the same calendar be reused across multiple date-based analyses
- Is the standard, recommended approach in Power BI modeling

Sort By Column

The Month column (text: 'Jan', 'Feb'...) was sorted by Month Number, and Week Day sorted by Day Number. Without this, Power BI sorts text alphabetically (April before January), breaking the visual timeline.

Mark as Date Table

The table was explicitly marked as a Date Table (with Date as the key column). This enables Power BI's built-in time-intelligence functions (DATEADD, TOTALYTD) to work correctly and builds a proper date hierarchy for drill-down.

6. DAX Measures

All measures are organized in a dedicated **Measures** table — best practice, keeps all KPIs discoverable in one place instead of scattered across data tables.

Measure	DAX	Purpose
Total Transaction Value	SUM(All_Transactions[Amount])	Total value processed

Total Transactions	COUNT(All_Transactions[Transaction_ID])	Total transaction count
Success Rate	DIVIDE([Successful Transactions],[Total Transactions])	% transactions succeeded
Total Users	DISTINCTCOUNT(All_Users[User_ID])	Unique user count
Trans Value MOM%	DIVIDE(current - previous, previous)	MoM growth in value
Total Trans MOM%	Same pattern, on transaction count	MoM growth in volume

Calculated Column vs Measure

A **calculated column** is computed row-by-row and stored physically in the table (e.g. Age Segment) — used when the result needs to be a filterable/sliceable field. A **Measure** is computed on the fly based on the current filter context (e.g. Total Transaction Value) — used for aggregations that must respond dynamically to slicers/filters. Measures are lighter on memory since they're not pre-computed and stored.

7. Advanced Power BI Techniques Used

a) Dynamic Conditional Formatting (Red/Green MoM)

The MoM % cards use rule-based conditional formatting: Value $\geq 0 \rightarrow$ Green, Value $< 0 \rightarrow$ Red. This makes growth/decline instantly visible without reading numbers — turning a static KPI into a decision-support signal.

b) Gradient Color-Coded Charts

Bar charts (Service revenue, Top Users) use gradient conditional formatting based on Total Transaction Value — low values shaded light purple (#CDB6F7), high values dark purple (#6D44C8). This visually ranks categories without needing a legend.

c) Dynamic Text (Natural Language Insights)

Two text boxes use Power BI's Q&A-style dynamic value insertion — e.g. '[Service] service gives the highest transaction value' — where [Service] is a live field reference that updates automatically as slicers change. This keeps the Insights section always accurate, even after filtering.

d) Drill-through Tooltips

Two custom tooltip pages were built: Tooltip 1 (Age Segment breakdown of transaction value) and Tooltip 2 (Service Type sub-category breakdown, e.g. which specific loan type drives the most value). This adds depth without cluttering the main dashboard.

8. Dashboard Design

- Custom background designed in Figma (not a default Power BI theme) — imported via Canvas Settings
- All default visual backgrounds/borders removed for a clean, card-based look
- KPI cards include embedded mini line/area charts for quick trend context
- Slicers styled as tiles (Month, Payment Status) for a modern filter UX

Design principle: kept intentionally simple — 4 KPIs, 2 slicers, 6 core visuals, 2 tooltips — rather than overloading it with every possible chart. A dashboard that's easy to explain in an interview is more valuable than one that's visually dense but hard to walk through.

9. Visual-by-Visual Insights & Recommendations

Visual	Insight	Recommendation
Transactions Over Time	Clear monthly seasonality — visible peaks and dips	Align marketing pushes with high-transaction months
Visual	Insight	Recommendation
Revenue by Service	Loans contribute the highest transaction value	Prioritize promoting loan products/offers in-app
Age Segment Contribution	Gen Z (20.7%) + Millennials (37.3%) = ~58% of users	Design campaigns targeting younger demographics
Weekday vs Weekend	71.6% of transactions on weekdays vs 28.4% weekend	Weekday-timed reminders likely to perform better
Top 5 Users	Small set of users drive disproportionate value	Consider loyalty rewards/VIP perks to retain them
Success Rate (96%)	4% fail due to Insufficient Amount/Server Error/Wrong PIN	Investigate server reliability; add low-balance alerts

10. Future Scope

While the current dashboard covers the core transactional analysis, potential future enhancements identified include:

- **Customer lifecycle segmentation** (New / Active / Dormant users based on join and last-transaction recency) — to support retention strategy
- **Failure reason deep-dive** — quantifying revenue lost per failure category to prioritize technical fixes

These extensions would shift the dashboard from purely descriptive ('what happened') toward more diagnostic, retention-focused analysis.

11. Tech Stack Summary

Power BI Desktop · Power Query (M) · DAX · Data Modeling (Star Schema) · Figma (UI Design) · Excel (Source Data)